

Scientific Machine Learning (SciML)

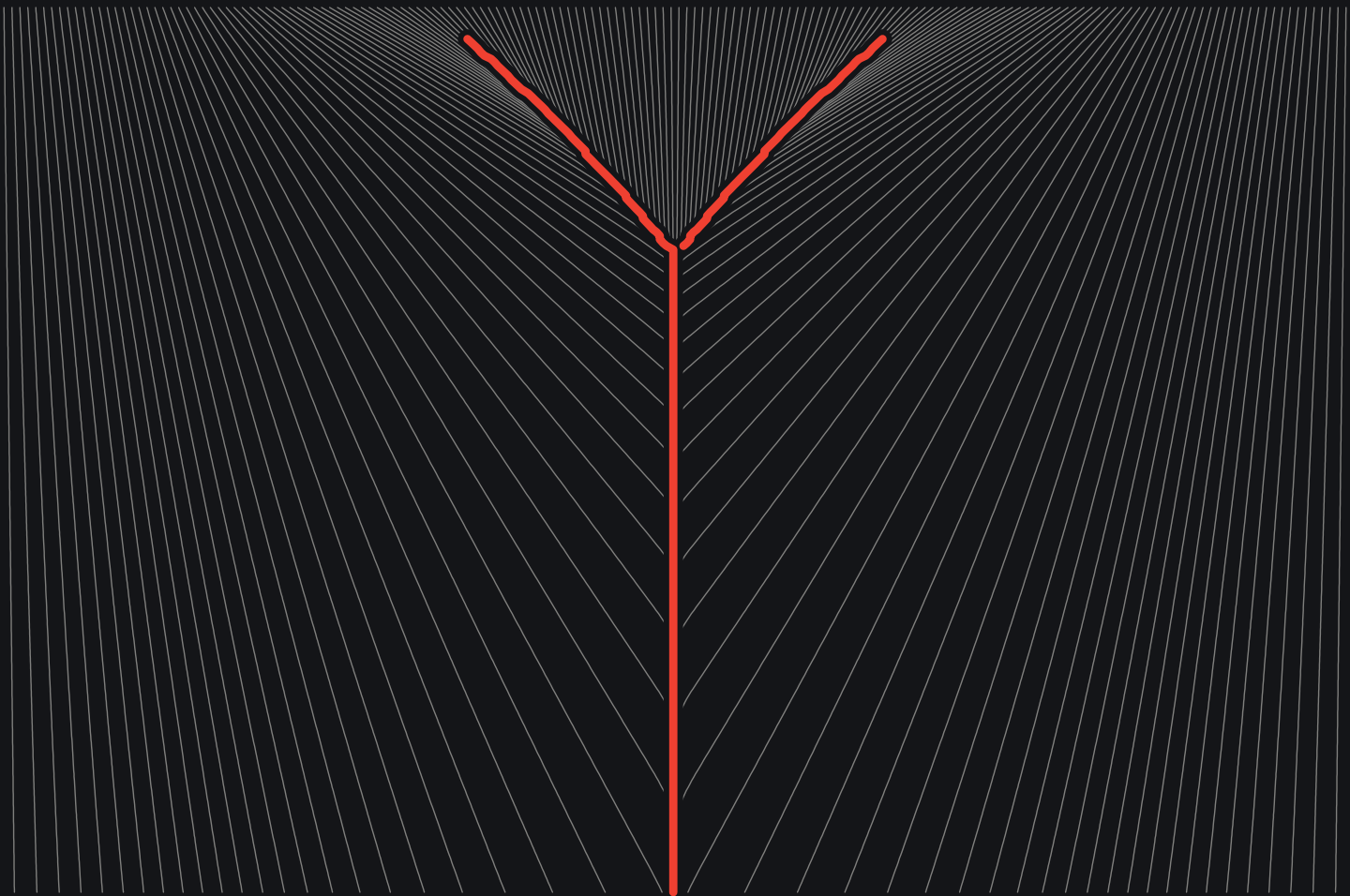


FIG. 0 Burgers' equation, $u_t + u \cdot u_x = (0.01/\pi) \cdot u_{xx}$, solved in this page by finite volumes (400 cells, 984 steps) from a perturbed start, $u_0 = -\sin(\pi \cdot x) + 0.39 \sin(3\pi \cdot x) - 0.12 \sin(5\pi \cdot x)$, to $t = 1.5$. Lines: 170 particles carried by the flow, time running down; red: shocks, where they collide.